

DATA ANALYSIS USING PYTHON-Internship

Task 4 (Medium) - Trend Analysis by Release Year

#Step_1.

```
# Step 1: Organize content by release year.
df=pd.read_csv("cleaned_netflix_titles.csv")

# trying to filterout the old dated movie but i provide you the first movie and TV show.
modern_df=df[df['release_year']>=2000]
organized_df=modern_df.groupby(['release_year','type']).size()
print("Organize content by release year")
print(organized_df.head(10))

print("-----")
print("The first movie and Tv show:")
first_df=df.groupby('type').agg({'release_year':'min','title':'first'})
first_df
```

Organize content by release year

release_year	type	
2000	Movie	33
	Tv Show	4
2001	Movie	40
	Tv Show	5
2002	Movie	44
	Tv Show	7
2003	Movie	51
	Tv Show	8
2004	Movie	55
	Tv Show	9

dtype: int64

The first movie and Tv show:

	release_year	title
type		
Movie	1942	Dick Johnson Is Dead
Tv Show	1925	Ganglands

Provided content data after year 2000. It helps to filter the new generation content.

To reminder I have provided first movie and TV show

Movie is Dick Johnson Is Dead and

TV Show is Ganglands.

#Step-2.

```
# Step 2: Calculate yearly content releases.

# Filter for modern titles (Year 2000 to present) to keep data focused
modern_df = df[df['release_year'] >= 2000]
yearly_content_df=modern_df.groupby(['release_year', 'type']).size().unstack()
yearly_content_df.head(10)
```

	type	Movie	Tv Show
release_year			
	2000	33	4
	2001	40	5
	2002	44	7
	2003	51	8
	2004	55	9
	2005	67	13
	2006	82	14
	2007	74	14
	2008	113	22
	2009	118	34

#Step-3.

```
# Step 3: Identify growth and decline trends.

# -> to find the growth and decline trend i need to evaluated yealy to yealry growing rate of movie and tv shows
#      so used pct_change() function to calculate the change from one year to the next.

growth_trend=yearly_content_df.pct_change() *100
print("--- Year-over-Year Growth/Decline Trends (%) ---")
print(growth_trend.tail(10).round(2))
```

```
--- Year-over-Year Growth/Decline Trends (%) ---
type      Movie  Tv Show
release_year
2012      19.31   57.50
2013      30.06   -3.17
2014      17.33   44.26
2015      50.00   80.68
2016      66.16   52.83
2017      16.26    9.05
2018       0.26   43.02
2019     -17.47    4.75
2020     -18.33    9.82
2021     -46.42  -27.75
```

The year 2015 and 2016 represent the most explosive period of growth in Netflix's history for both formats.

In 2015, Movie output grew by 50.00% and TV Shows skyrocketed by 80.68%.

This velocity continued into 2016 with Movies peaking at massive 66.16% growth rate and TV shows decline to 52.83% but this is good growth for Netflix.

The Great Strategy Pivot (2013):

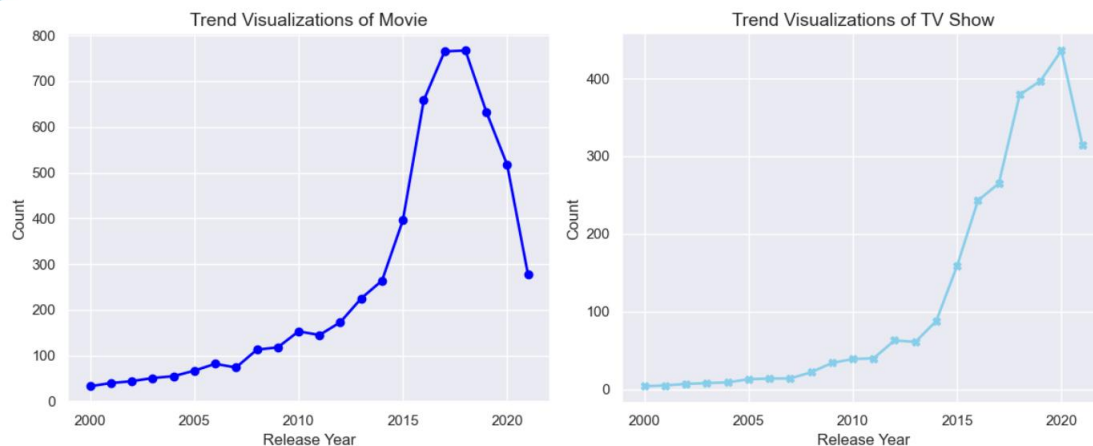
->In 2013, Movie production grew significantly (30.06%), while TV Shows experienced a clear decline of -3.17%. This represents a brief historical moment where Netflix focused on acquiring massive Hollywood movie licensing blocks before shifting its long-term financial focus heavily back into serialized multi-season series.
in next year 2014 clearly see TV Shows growth by 44.26%.

#Step-4.

```
# Step 4: Create charts and rankings.
```

```
sns.set_theme()
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(yearly_content_df.index, yearly_content_df['Movie'], color='blue', marker='o', linewidth=2)
plt.title("Trend Visualizations of Movie", fontsize=14)
plt.xlabel("Release Year")
plt.ylabel("Count")

# plot for TV show
plt.subplot(1, 2, 2)
plt.plot(yearly_content_df.index, yearly_content_df['TV Show'], color='skyblue', marker='x', linewidth=2)
plt.title("Trend Visualizations of TV Show", fontsize=14)
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



Production Aggression Era (2015–2020): Netflix experienced a massive surge in inventory expansion during this window. **Movie additions peaked sharply between 2017–2018** (reaching nearly 800 annual titles), while **TV Show assets peaked slightly later in 2020** (reaching over 400 annual titles).

step 5:Generate business insights.

While both mediums exhibit upward trajectories, their operational velocity and growth curves differ significantly.

The Pattern: Movie production volumes scale aggressively, showing steeper vertical spikes in short cycles. Conversely, TV Show growth is more deliberate, incremental, and stable year-over-year. Sharp, sudden drops or plateaus in both line graphs during identical chronological windows (e.g., historical periods like the 2007–2008 financial crisis, the 2020 global pandemic, or major industry labor strikes).

Movies are transaction and high-risk/high-reward:

-> Production companies aggressively ramp up output to diversify their portfolio. If one blockbuster fails, three others might cover the costs.

TV Shows are relational asset engines.

-> A television series requires sustained, multi-year financial commitments (renewals, recurring talent contracts).

